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Seeding Density Outweighs Light Intensity in Microgreens Yield on a Vertical Farming System

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Highlights

What are the main findings?

- In all species investigated, SD was more impactful on microgreens' fresh and dry weight than PPFD;
- Mineral content in sunflower microgreens was not affected by PPFD or SD, likely due to sufficient seed reserves;
- The anthocyanin content in red cabbage microgreens increased under a higher SD, unlike sunflower, radish, and green cabbage.

What are the implications of the main findings?

- Farmers and technicians are encouraged to prioritize adjusting SD in microgreens production, as a PPFD of 50 µmol m^{-2} s^{-1} was enough to meet plant demands in sunflower, radish, red cabbage, and, to a lesser extent, green cabbage.

Abstract

The optimal photosynthetic photon flux density (PPFD) and seeding density (SD) combinations for microgreens production on vertical farming systems remain unclear for many species. This study aimed to determine the effects of PPFD (50, 100, 150, and 200 µmol m^{-2} s^{-1}) and SD for sunflower (1275, 1700, and 2125 g m^{-2}), radish (272, 340, and 408 g m^{-2}), and green and red cabbage (126, 180, and 235 g m^{-2}) microgreens. PPFD had no effect on the yield of sunflower, radish, or red cabbage, indicating that 50 µmol m^{-2} s^{-1} was sufficient to meet plant demands. Green cabbage yield increased ~10% at 150 µmol m^{-2} s^{-1}. Increasing SD enhanced yields, with gains of up to ~22%, 49%, 52%, and 30% for sunflower, radish, green, and red cabbage, respectively (1700, 408, 235, and 180 g m^{-2}). PPFD did not affect the nutrient content of sunflower, likely due to its high seed reserves. A higher SD increased anthocyanin in red cabbage but diminished its content in all other species. Our findings highlight that SD is more impactful than PPFD on determining yield, at least

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under low irradiance, in sunflower, radish, and red cabbage microgreens grown on indoor vertical farming systems.

Keywords: artificial lighting; photosynthetic photon flux density (PPFD); seed density; phytochemical quality; mineral content; plant factories

1. Introduction

Microgreens are superfoods obtained by a complete shift in the way traditional crops are grown. The methods for microgreen production emphasize a dense plant population that is harvested while in the seedling stage, between the full development of the cotyledons and the emergence of the first true leaf, at a height of approximately 3–9 cm. Thus, the production cycle ranges from 7 to 28 days [1,2]. A major benefit of microgreens is their inherently high content of bioactive compounds, given that seed reserves, in combination with proper fertilization, enable the concentration of anthocyanins, carotenoids, and many other metabolites in a small biomass volume. As a result, fresh, palatable, and highly nutritious food is obtained, and market interest in microgreens production has increased in the past decade [3].

The Brassicaceae family is widely used for microgreen production because its species exhibit rapid germination, short growth cycles, and considerable variation in color, flavor, and morphology, in addition to high concentrations of compounds beneficial to human health [4]. Among the species commonly cultivated as microgreens are radish (*Raphanus sativus* L.) and green and red cabbage (*Brassica oleracea* L.). Sunflower (*Helianthus annuus* L.), though not commonly used as a fresh food in its mature stage, can be grown as a microgreen and has a strong market interest due to its high yield potential.

Microgreen production, due to its technical definition, requires high seeding densities (SD) in which seeds are sown with little to no spacing between them [5]. SD varies according to seed size and the developmental stage at which the crop is intended to be harvested [6,7]. Increasing SDs initially lead to higher yields, though at a loss in individual seedling mass and height [8]. This reduction eventually outweighs overall productivity, hindering efficiency. Conversely, at higher SD, seedlings become taller and develop smaller cotyledons, and depending on market preference, one trait may be prioritized over another [9]. An appropriate SD is, therefore, essential to achieve a satisfactory yield and quality while also representing a significant component of production costs [10].

In response to global climate challenges and with the advent of the light-emitting diode (LED) technology, controlled-environment agriculture using artificial lighting has gained increasing attention worldwide. Among its advantages, LEDs can be positioned close to the crop canopy because of their relatively low radiant heat emission, allowing a greater utilization of the emitted radiation (photons), thus facilitating verticalization [11]. Consequently, productivity can be increased by exploiting the third dimension of cultivation through the vertical stacking of production layers, with the addition of minimal external weather interference and the control of ambient air temperature and humidity. Given the potential for high productivity in a shorter production cycle, these vertical farming systems are commonly referred to as plant factories [12].

Light intensity, in this context, is a measurement of the photosynthetic photon flux density (PPFD), or the irradiance that is used by plants in their photosynthetic processes. Since PPFD is proportional to the energy expenditure, it is considered one of the governing factors concerning the commercial viability of vertical farming systems. Microgreens are the most efficient crop in plant factories in terms of photon requirement in relation to

dry biomass production and market value [13], since light demands are lower during the seedling stage—which is also when the plants are harvested. Several studies have demonstrated that PPFD affects microgreen productivity and quality, with responses varying among species [6,14–20]. There is a focus on achieving a balance between irradiance and yield, since a low photon-use efficiency may result in unnecessary infrastructure and energy costs, thereby affecting the economic viability of plant factories.

As an evolving subject, strategies concerning the optimal combinations of PPFD and SD remain unclear, as these factors are often investigated individually. In this study, we hypothesized that increasing PPFD and SD would interact and improve sunflower, radish, and green and red cabbage microgreens yield and quality, and investigated their effects on yield, biometry, mineral content, and phytochemical quality.

2. Materials and Methods

2.1. Plant Material and Growth Conditions

Four independent experiments were carried out at the Indoor Farming Laboratory (Agronomic Institute IAC, Campinas, SP, Brazil), where air temperature was maintained at 25 ± 1 °C with air conditioning. Non-treated seeds (Agristar do Brasil, Santo Antônio de Posse, SP, Brazil) of sunflower (*Helianthus annuus* L. var. Miúdo), radish (*Raphanus sativus* L. var. Crimson Gigante) and green (*Brassica oleracea* L. var. capitata Chato de Quintal) and red (*Brassica oleracea* L. var. capitata f. rubra) cabbage were sourced from a local market.

Microgreens were sown on pairs of plastic trays (580 × 280 × 30 mm, JKS Agro, Guarulhos, SP, Brazil). Each pair contained one tray with openings on the bottom to facilitate irrigation through capillarity, and another without openings to keep irrigation water. The former was filled with a 2 cm layer of coconut fiber substrate (Golden Mix 11, pH 6.0, 14-16-18 NPK, Amafibra, Artur Nogueira, SP, Brazil).

Cultivation was separated into dark and light stages: firstly, trays were stacked in groups of four and kept in the dark for germination and initial shoot and root development. Another tray filled with substrate was placed on top of each stack to provide enough weight to induce uniform plant establishment. After four days, trays were unstacked and placed on multi-layered growing racks (0.9 × 0.6 m; 0.28 m height of each layer) for the light stage, which lasted until harvest.

Plants were irrigated daily from the start of the light stage until the day before the harvest with nutrient solution [21] containing: N-NO_3: 0.185; N-NH_4: 0.018; P: 0.0915; K: 0.225; Ca: 0.1425; Mg: 0.036 and S: 0.044 for macronutrients (g L^{-1}); and Fe: 1.815; Mn: 0.455; Cu: 0.455; Zn: 0.182; B: 0.455 and Mo: 0.09 for micronutrients (mg L^{-1}). Nutrient solution electrical conductivity was 1.7 mS cm^{-1}, and pH was adjusted to 6.5 with ortho-phosphoric acid. No EC or pH adjustments were needed.

Sunflower microgreens were harvested when the first true leaf had reached approximately 2 mm in length. For every other species, harvest occurred when cotyledons were expanded. Regardless of species, plants were harvested manually with scissors ~1 cm above the substrate level 10 days after sowing.

2.2. Light Intensity and Seed Density Treatments

Each layer of the growing rack contained a set of LED lamps (GLP-v.11.7.1-21, LEDsUp, Botucatu, SP, Brazil), whose spectra peaked at 450 nm (blue, 26%) and 640 nm (red, 57%). Four light intensity—PPFD—treatments were defined by the number of LED lamps installed in each layer, at a canopy height of ~0.25 m: 50, 100, 150 and 200 µmol m^{-2} s^{-1}, respectively 6, 10, 14 and 18 lamps, measured with a quantometer (Li 250-A, LI-COR, Lincoln, NE, USA; 16 samples per layer with <10% variation). Photoperiod on the light stage was fixed at 16 h.

Seed density (SD) treatments were defined individually for each species. Sunflower microgreens SD was 1275, 1700 and 2125 g m^{-2}, radish SD was 272, 340 and 408 g m^{-2}, and green and red-cabbage SD was 126, 180 and 235 g m^{-2}.

2.3. Yield, Biometric Characteristics and Mineral Content

The following production variables were evaluated: (a) shoot fresh weight (FW, kg m^{-2}), corresponding to the total fresh biomass of all microgreens harvested from a tray adjusted to m^2; (b) shoot dry weight (DW, g m^{-2}), determined after drying in a forced-air oven at 60 °C until constant weight; and (c) fresh yield-to-seed (Y/S) ratio, calculated by dividing the FW by SD. Dry shoot samples were then analyzed for macro (N, P, K, Ca, Mg, S) and micronutrient (Fe, Cu, Mn, B, Zn) content at the Soil Analysis Laboratory (Agronomic Institute IAC, Campinas, SP, Brazil).

Microgreen morphological traits were measured using a digital caliper (500-196-30 Absolute, Mitutoyo, Kanagawa, Japan): (a) hypocotyl length (HL, mm), measured from the cutting point (harvest point) to the apical meristem; (b) cotyledon longitudinal length (CLL, mm), measured from the base of the cotyledon (petiole insertion at the meristem) to the cotyledon apex; and (c) cotyledon transversal length (CTL, mm), measured across the widest portion of the cotyledon.

2.4. Phytochemical Traits

Phytochemical quality was evaluated after harvest by determining: (a) pH, measured in 100 mL of an aqueous extract [22]; (b) soluble solids content (SS, °Brix), measured using a digital refractometer (Pocket Refractometer PAL-1, Atago Corporation, Niigata, Japan); (c) titratable acidity (TA, mg citric acid 100 g^{-1}) [22]; and (d) chlorophyll a (Chl a), chlorophyll b (Chl b), carotenoid (CA), and anthocyanin (AN) contents, with adaptations (mg 100 g^{-1}) [23].

2.5. Experimental Design and Statistical Analyses

Each species was treated as an independent experiment, and therefore no direct comparison was analyzed statistically. A split-plot design was defined with 4 PPFD × 3 SD treatments and 3 replicates. Each tray represented an experimental unit, and hypocotyl and cotyledon characteristics were obtained by averaging three individual plants per tray. Values expressed in m^2 were calculated from the tray area (0.1652 m^2).

Data were analyzed statistically using ANOVA to assess the interaction and isolated effects of PPFD and SD treatments. Regression analysis was performed whenever PPFD treatments differed significantly at $p < 0.05$, and Tukey's means test was performed whenever SD treatments differed significantly, also at $p < 0.05$. Data processing was performed on SISVAR 5.6 (UFLA, Lavras, MG, Brazil), and when significance was found in regression analyses, graphs were plotted on OriginPro 8.5 (OriginLab Corporation, Northampton, MA, USA).

3. Results

3.1. Yield and Biometric Characteristics

No interaction was observed between PPFD and SD treatments. Fresh (FW) and dry weight (DW) were affected by PPFD only in green cabbage microgreens. Increasing PPFD increased FW and DW up to an estimated maximum of 3.8 kg m^{-2} at 158 µmol m^{-2} s^{-1}, and 21.9 g m^{-2} at 171 µmol m^{-2} s^{-1} (Figure 1). In sunflower, radish, and red cabbage microgreens, PPFD had no significant main effect. Mean FW values were 7.1, 4.4, and 2.7 kg m^{-2}, respectively, while DW averaged 75, 181, and 158 g m^{-2}.

Regarding morphology, increasing PPFD had a significant independent effect only on the sunflower microgreens experiment, increasing cotyledon transversal (CTL) and

longitudinal (CLL) length by 1.7 and 3.3 mm, respectively (Figure 2). No significant effect of PPFD was observed on CTL or on CLL in radish, green cabbage, or red cabbage, with mean values of 18, 8, and 8 mm for CTL and 25, 11, and 15 mm for CLL, respectively. These results indicate that the response to increasing PPFD varies among species, and, among the species evaluated, only sunflower exhibited greater cotyledon expansion in response to the higher incident radiation. Hypocotyl length did not differ in relation to PPFD for any species, with averages for sunflower, radish, green and red cabbage of 58, 60.7, 68.1 and 58.9 mm.

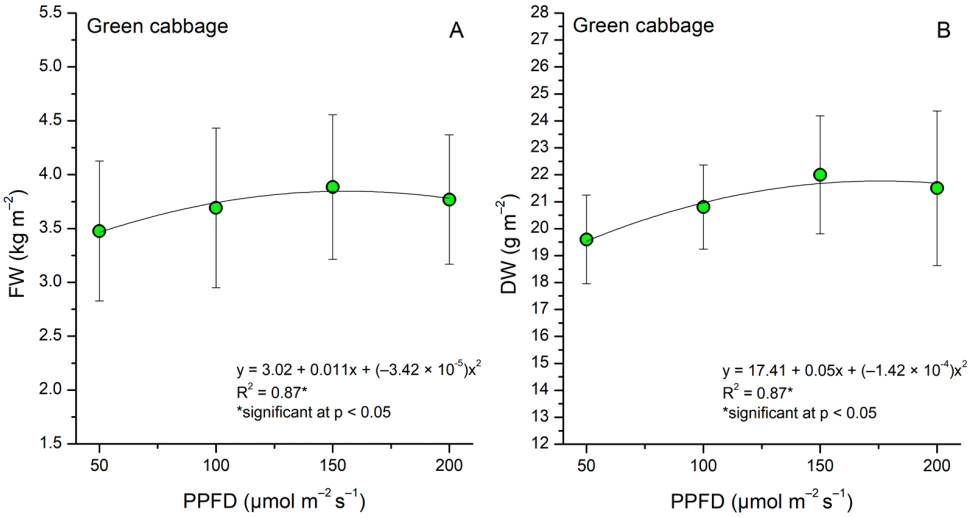


Figure 1. Shoot fresh (FW) (**A**) and dry (DW) (**B**) weight of green cabbage microgreens as a function of light intensity (PPFD).

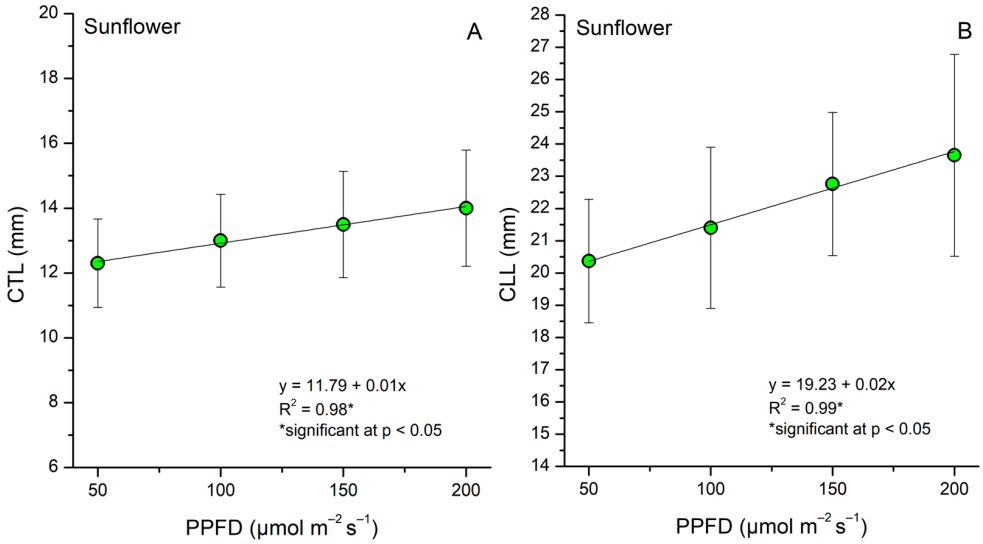


Figure 2. Cotyledon transversal (CTL) (**A**) and longitudinal (CLL) (**B**) length of sunflower microgreens as a function of light intensity (PPFD).

Whereas PPFD, within the range evaluated, affected relatively few response variables, seeding density (SD) exerted significant independent effects on most of the measured characteristics across all species studied, including morphological traits, yield, mineral and physicochemical characteristics. These results indicate that, under the conditions of this study, seeding density had a greater influence on crop performance than PPFD. The averages for shoot fresh (FW) and dry (DW) weight, yield-to-seed ratio (Y/S), hypocotyl length (HL), and cotyledon transversal (CTL) and longitudinal (CLL) length of microgreens in relation to SD are presented in Table 1.

Table 1. Shoot fresh (FW) and dry (DW) weight, yield-to-seed ratio (Y/S), hypocotyl length (HL) and cotyledon transversal (CTL) and longitudinal (CLL) length of sunflower, radish, green and red cabbage microgreens in function of seeding density (SD).

SD	FW	DW	Y/S	HL	CTL	CLL
g m^{-2}	kg m^{-2}	g m^{-2}	-	---------- mm ----------		
			Sunflower			
1275	6.5 ± 0.55 c [1]	67.9 ± 4.47 b	5.1 ± 0.4 a	65 ± 9.4 a	14 ± 1.1 a	22 ± 2.2 a
1700	7.9 ± 0.58 a	67.2 ± 6.34 b	4.6 ± 0.4 b	61.2 ± 5.7 a	14 ± 1.0 a	23 ± 2.4 a
2125	7.4 ± 0.89 b	88.9 ± 9.52 a	3.5 ± 0.4 c	48 ± 4.3 b	11 ± 1.2 b	19 ± 1.9 b
			Radish			
272	3.5 ± 0.47 c	150.2 ± 36.73 c	12.9 ± 1.8 a	59.3 ± 6.6 a	19 ± 0.9 a	35 ± 1.8 a
340	4.3 ± 0.52 b	182.6 ± 33.81 b	12.6 ± 1.6 a	65.0 ± 9.3 a	18 ± 0.7 a	30 ± 2.3 b
408	5.2 ± 0.51 a	210.0 ± 36.04 a	12.7 ± 1.8 a	57.9 ± 10.1 a	15 ± 1.2 b	30 ± 2.0 b
			Green cabbage			
126	2.9 ± 0.21 c	107.0 ± 10.6 c	23.0 ± 1.7 a	67.2 ± 8.0 a	8.4 ± 0.83 a	16 ± 2.1 a
180	3.7 ± 0.20 b	130.7 ± 8.3 b	20.6 ± 1.1 b	67.4 ± 4.9 a	7.8 ± 0.68 b	14 ± 1.1 b
235	4.4 ± 0.24 a	158.4 ± 9.2 a	18.7 ± 1.0 c	69.2 ± 4.7 a	7.2 ± 0.67 c	13 ± 1.0 b
			Red cabbage			
126	2.3 ± 0.34 b	112.1 ± 38.80 c	18.3 ± 2.3 a	61.0 ± 5.6 a	9.1 ± 1.50 a	16 ± 0.9 a
180	3.0 ± 0.46 a	155.7 ± 46.49 b	16.7 ± 1.9 a	57.2 ± 9.5 a	8.4 ± 1.81 b	14 ± 0.8 b
235	3.0 ± 0.80 a	204.9 ± 62.38 a	12.8 ± 2.1 b	56.2 ± 7.2 a	8.0 ± 1.41 b	14 ± 1.1 b

[1] Means in the same column that are followed by the same letter do not differ from one another according to Tukey's test ($p > 0.05$).

3.2. Mineral Content

No interaction was observed between PPFD and SD treatments regarding nutrient content, but we detected isolated effect instances in all species tested.

Tables 2 and 3 present the averages for macronutrient or micronutrient content under increasing PPFD. The light treatments influenced macronutrient content in radish, green and red cabbage. Regarding N, the light increases reduced content by up to 11 and 18% under 200 µmol m^{-2} s^{-1} for radish and red cabbage. This decrease also happened for P, Mn, and Zn for red cabbage and K for green cabbage. The opposite was observed for P, K, and Ca for radish, and K for red cabbage.

Table 2. Nitrogen (N), phosphorus (P), potassium (K), calcium (Ca), magnesium (Mg), and sulfur (S) contents in the shoot (hypocotyl and cotyledon leaves) of sunflower, radish, green and red cabbage microgreens in function of light intensity (PPFD).

PPFD	N	P	K	Ca	Mg	S
µmol m^{-2} s^{-1}	---------- g kg^{-1} ----------					
			Sunflower			
50	48.0 ± 0.8 a [1]	10.4 ± 1.4 a	25.4 ± 2.3 a	3.9 ± 0.7 a	5.7 ± 0.7 a	5.6 ± 0.9 a
100	46.8 ± 1.2 a	10.2 ± 0.5 a	23.2 ± 1.8 a	3.7 ± 0.4 a	5.5 ± 0.3 a	5.8 ± 0.3 a
150	48.4 ± 2.0 a	11.0 ± 0.9 a	24.0 ± 2.5 a	4.0 ± 0.7 a	5.9 ± 0.4 a	6.3± 0.8 a
200	46.8 ± 4.0 a	10.5 ± 0.6 a	23.9 ± 1.5 a	3.6 ± 0.6 a	5.8 ± 0.3 a	6.1 ± 0.5 a
			Radish			
50	52.3 ± 2.4 a	12.2 ± 0.5 b	39.0 ± 3.7 b	10.5 ± 1.2 a	5.8 ± 0.3 a	18.7 ± 0.9 b
100	48.6 ± 2.2 b	12.5 ± 0.4 b	50.5 ± 5.5 a	10.9 ± 1.0 a	5.7 ± 0.3 a	20.1 ± 1.2 b
150	47.6 ± 2.5 b	13.2 ± 1.0 a	53.2 ± 3.7 a	10.9 ± 1.0 a	5.9 ± 0.4 a	22.0 ± 1.7 a
200	46.6 ± 1.6 b	13.0 ± 1.2 a	52.4 ± 2.4 a	10.4 ± 1.1 a	5.7 ± 0.5 a	22.0 ± 2.0 a

Table 2. *Cont.*

PPFD	N	P	K	Ca	Mg	S
µmol m^{-2} s^{-1}	g kg^{-1}					
			Green cabbage			
50	63.7 ± 4.2 b	14.8 ± 1.4 a	48.1 ± 3.9 a	8.6 ± 0.8 a	4.4 ± 0.3 a	26.2 ± 3.6 a
100	66.7 ± 2.4 a	14.6 ± 2.0 a	38.1 ± 3.5 b	8.2 ± 1.1 a	4.4 ± 0.4 a	26.6 ± 3.1 a
150	63.8 ± 3.0 b	14.4 ± 1.5 a	39.1 ± 4.0 b	8.0 ± 0.8 a	4.4 ± 0.3 a	26.4 ± 3.3 a
200	62.7 ± 3.3 b	13.8 ± 1.4 a	37.6 ± 4.8 b	7.6 ± 0.9 a	4.2 ± 0.3 a	24.5 ± 2.8 a
			Red cabbage			
50	50.9 ± 2.2 a	11.4 ± 0.8 a	39.7 ± 3.0 b	10.7 ± 1.0 a	4.8 ± 0.4 ab	24.2 ± 2.0 a
100	47.0 ± 2.0 b	10.9 ± 0.8 a	48.2 ± 7.7 a	11.2 ± 1.5 a	5.1 ± 0.3 ab	24.5 ± 2.7 a
150	45.6 ± 2.9 b	10.7 ± 0.7 a	47.9 ± 2.4 a	11.3 ± 1.6 a	5.3 ± 0.4 a	23.4 ± 1.7 a
200	41.7 ± 2.7 c	9.2 ± 1.0 b	41.7 ± 6.0 b	11.0 ± 1.0 a	4.6 ± 0.5 b	19.0 ± 2.5 b

[1] Means in the same column that are followed by the same letter do not differ from one another according to Tukey's test ($p > 0.05$).

Table 3. Boron (B), copper (Cu), iron (Fe), manganese (Mn), and zinc (Zn) contents in the shoot (hypocotyl and cotyledon leaves) of sunflower, radish, green and red cabbage microgreens in function of light intensity (PPFD).

PPFD	B	Cu	Fe	Mn	Zn
µmol m^{-2} s^{-1}	mg kg^{-1}				
			Sunflower		
50	26.1 ± 5.3 a [1]	29.2 ± 4.1 a	112.0 ± 13.6 a	57.9 ± 6.2 a	98.5 ± 13.1 a
100	25.8 ± 2.1 a	25.7 ± 2.0 a	105.0 ± 4.5 a	69.2 ± 3.6 a	95.9 ± 5.8 a
150	28.3 ± 5.8 a	27.9 ± 2.2 a	111.0 ± 8.7 a	64.9 ± 4.9 a	104.3 ± 8.1 a
200	24.1 ± 3.9 a	26.5 ± 1.3 a	102.2 ± 8.1 a	64.0 ± 5.4 a	99.6 ± 3.9 a
			Radish		
50	42.9 ± 4.9 a	5.9 ± 0.4 a	95.5 ± 13.2 a	66.9 ± 6.0 a	68.8 ± 6.5 a
100	46.6 ± 3.8 a	5.8 ± 1.2 a	97.0 ± 27.5 a	78.1 ± 6.5 a	68.2 ± 3.3 a
150	51.7 ± 5.4 a	6.5 ± 0.6 a	89.0 ± 13.3 a	70.1 ± 8.8 a	70.7 ± 7.0 a
200	50.1 ± 7.6 a	6.1 ± 0.8 a	90.2 ± 18.2 a	71.3 ± 7.5 a	72.3 ± 10.0 a
			Green cabbage		
50	40.3 ± 3.3 a	8.6 ± 1.4 a	203.9 ± 29.7 a	63.6 ± 4.8 a	135.8 ± 17.5 a
100	37.7 ± 5.1 a	7.8 ± 1.5 a	178.4 ± 18.5 a	62.5 ± 4.8 a	126.5 ± 21.3 a
150	37.7 ± 3.7 a	8.1 ± 3.7 a	192.2 ± 31.4 a	59.3 ± 3.2 a	124.2 ± 15.3 a
200	36.9 ± 5.2 a	8.0 ± 5.2 a	190.5 ± 31.2 a	63.2 ± 3.0 a	121.1 ± 19.4 a
			Red cabbage		
50	37.6 ± 4.0 a	5.6 ± 0.6 a	114.0 ± 17.5 a	47.5 ± 7.6 b	87.3 ± 12.5 a
100	46.6 ± 5.9 a	5.9 ± 0.6 a	119.7 ± 10.4 a	59.3 ± 10.6 a	85.5 ± 7.2 a
150	45.4 ± 5.3 a	5.8 ± 0.7 a	120.9 ± 17.7 a	59.6 ± 9.3 a	83.1 ± 8.2 a
200	40.5 ± 6.6 a	5.2 ± 0.1 a	102.4 ± 21.3 a	48.0 ± 8.1 b	69.4 ± 8.9 b

[1] Means in the same column that are followed by the same letter do not differ from one another according to Tukey's test ($p > 0.05$).

Micronutrient contents (B, Cu, Fe, Mn, Zn) for all species were also not influenced by PPFD, except for Mn and Zn in red cabbage.

The averages for nutrient content under different seeding densities (SD) are presented in Tables 4 and 5. Notably, SD have not influenced radish nutrient content, except for Zn, which had a higher content under 408 g m^{-2}. Conversely, a pronounced effect was observed in green cabbage macronutrients, where only N content stayed the same between SD treatments.

Table 4. Nitrogen (N), phosphorus (P), potassium (K), calcium (Ca), magnesium (Mg), and sulfur (S) contents in the shoot (hypocotyl and cotyledon leaves) of sunflower, radish, green and red cabbage microgreens in function of seeding density (SD).

SD	N	P	K	Ca	Mg	S
$g\ m^{-2}$	---------- $g\ kg^{-1}$ ----------					
			Sunflower			
1275	47.7 ± 1.7 a [1]	11.0 ± 0.8 a	24.7 ± 1.9 a	3.8 ± 0.4 ab	5.9 ± 0.5 a	6.2 ± 0.5 a
1700	48.0 ± 3.6 a	10.4 ± 1.1 ab	24.6 ± 2.4 a	4.2 ± 0.7 a	5.6 ± 0.5 a	6.2 ± 0.7 a
2125	48.3 ± 1.5 a	10.2 ± 0.7 b	23.0 ± 1.9 a	3.4 ± 0.5 b	5.7 ± 0.4 a	5.4 ± 0.5 b
			Radish			
272	47.7 ± 2.5 a	12.4 ± 0.9 a	50.5 ± 7.9 a	11.0 ± 1.1 a	5.7 ± 0.4 a	20.7 ± 2.0 a
340	48.6 ± 2.7 a	13.0 ± 0.7 a	48.5 ± 7.7 a	10.5 ± 1.1 a	5.6 ± 0.2 a	21.0 ± 2.2 a
408	49.9 ± 3.5 a	12.8 ± 1.1 a	47.3 ± 6.3 a	10.5 ± 1.0 a	5.8 ± 0.5 a	20.5 ± 2.1 a
			Green cabbage			
126	62.7 ± 2.7 a	15.8 ± 0.9 a	43.7 ± 5.5 a	8.7 ± 0.9 a	4.5 ± 0.2 a	26.2 ± 3.7 a
180	64.9 ± 3.5 a	14.7 ± 1.1 b	39.6 ± 6.6 b	7.7 ± 0.8 b	4.2 ± 0.2 b	23.4 ± 1.9 b
235	64.9 ± 3.9 a	12.7 ± 0.7 c	38.8 ± 4.3 b	7.9 ± 0.9 b	4.3 ± 0.3 b	28.2 ± 1.7 a
			Red cabbage			
126	44.8 ± 3.5 b	10.6 ± 0.9 a	47.6 ± 5.7 a	11.2 ± 1.2 a	5.2 ± 0.4 a	23.3 ± 2.5 a
180	46.9 ± 4.3 a	10.7 ± 0.9 a	45.8 ± 4.5 a	11.6 ± 1.3 a	5.0 ± 0.6 a	22.9 ± 2.5 a
235	47.2 ± 4.3 a	10.3 ± 1.6 a	40.2 ± 6.9 b	10.1 ± 1.1 b	4.7 ± 0.4 b	22.0 ± 4.1 a

[1] Means in the same column that are followed by the same letter do not differ from one another according to Tukey's test ($p > 0.05$).

Table 5. Boron (B), copper (Cu), iron (Fe), manganese (Mn), and zinc (Zn) contents in the shoot (hypocotyl and cotyledon leaves) of sunflower, radish, green and red cabbage microgreens in function of seeding density (SD).

SD	B	Cu	Fe	Mn	Zn
$g\ m^{-2}$	---------- $mg\ kg^{-1}$ ----------				
			Sunflower		
1275	28.3 ± 3.6 a [1]	27.7 ± 2.8 a	109.7 ± 11.4 a	65.0 ± 4.3 a	104.2 ± 7.9 a
1700	27.5 ± 5.1 a	26.7 ± 3.8 a	107.5 ± 11.4 a	62.0 ± 6.5 a	97.0 ± 10.5 a
2125	22.4 ± 2.3 b	27.5 ± 1.6 a	105.4 ± 6.3 a	64.9 ± 7.9 a	97.5 ± 7.6 a
			Radish		
272	46.8 ± 6.1 a	5.9 ± 0.5 b	77.3 ± 8.1 a	67.5 ± 7.7 a	65.8 ± 1.7 b
340	48.3 ± 5.8 a	5.8 ± 1.1 b	97.8 ± 23.0 a	75.9 ± 5.6 a	68.8 ± 1.7 b
408	48.4 ± 7.4 a	6.5 ± 0.7 a	103.7 ± 10.7 a	71.8 ± 7.7 a	75.4 ± 1.7 a
			Green cabbage		
126	41.1 ± 4.1 a	8.9 ± 1.1 a	198.6 ± 30.1 a	63.3 ± 4.2 a	141.2 ± 11.2 a
180	36.2 ± 3.9 b	9.1 ± 0.9 a	202.2 ± 33.9 a	61.8 ± 3.5 a	132.4 ± 13.9 a
235	37.2 ± 3.8 b	6.4 ± 0.8 b	172.8 ± 27.2 a	61.4 ± 5.0 a	107.1 ± 9.9 b
			Red cabbage		
126	44.2 ± 5.7 a	5.4 ± 0.8 b	100.4 ± 11.1 b	53.2 ± 11.2 a	77.9 ± 7.8 a
180	44.0 ± 5.3 a	6.0 ± 0.8 a	122.2 ± 18.9 a	57.2 ± 10.6 a	84.1 ± 11.0 a
235	39.3 ± 7.4 a	5.4 ± 0.7 b	120.2 ± 15.4 a	50.5 ± 9.2 a	81.9 ± 14.7 a

[1] Means in the same column that are followed by the same letter do not differ from one another according to Tukey's test ($p > 0.05$).

3.3. Phytochemical Traits

No interaction was observed between PPFD and SD in the phytochemical quality in any of the species investigated. PPFD alone had no significant effect in any of the phytochemical parameters evaluated, where sunflower microgreens had, respectively, the following contents on average (mg 100 g^{-1}): 2.2 (Chlorophyll a, Chl a), 2.6 (chlorophyll b, Chl b), 0.7 (anthocyanins, AN), 0.7 (carotenoids, CA), 89 (titratable acidity, TA), 3 (soluble solids, SS) and 6.4 (pH). Radish: 3.6 (Chl a), 3.8 (Chl b), 1.0 (AN), 0.6 (CA), 134 (TA), 2 (SS) and 6.4 (pH). Green cabbage: 3.5 (Chl a), 3.9 (Chl b), 1.2 (AN), 0.7 (CA), 89 (TA), 2 (SS) and 6.6 (pH). Red cabbage: 3.4 (Chl a), 3.8 (Chl b), 3.8 (AN), 0.7 (CA), 101 (TA), 2 (SS) and 6.2 (pH).

In contrast, seeding density (SD) had a pronounced effect on the phytochemical composition of sunflower, radish, and green cabbage microgreens (Table 6). Increasing SD reduced Chl a concentration by 32 and 16%, respectively, in sunflower and radish, whereas green cabbage increased by 50%. AN concentrations also decreased with increasing SD in sunflower, radish, and green cabbage microgreens by 13, 10, and 21%, respectively. In contrast, red cabbage exhibited an increase of 62% in AN concentration at the highest SD. Likewise, increasing SD reduced CA concentrations in sunflower, radish, and green cabbage by 13, 29, and 44%, respectively, while no significant effect was detected in red cabbage.

Table 6. Chlorophyll a (Chl a), chlorophyll b (Chl b), anthocyanins (AN), carotenoids (CA), titratable acidity (TA), soluble solids (SS) and pH of microgreens in function of seeding density (SD).

SD	Chl a	Chl b	AN	CA	TA	SS	pH
g m^{-2}	**mg 100 g^{-1}**					**°Brix**	**---**
Sunflower							
1275	2.5 ± 0.4 a [1]	3.0 ± 0.4 a	0.8 ± 0.1 a	0.8 ± 0.1 a	87 ± 8.1 a	3 ± 0.4 b	6.4 ± 0.06 a
1700	2.4 ± 0.2 a	2.8 ± 0.3 a	0.8 ± 0.1 a	0.7 ± 0.1 b	87 ± 7.5 a	3 ± 0.1 b	6.4 ± 0.05 a
2125	1.7 ± 0.4 b	2.1 ± 0.4 b	0.7 ± 0.1 b	0.7 ± 0.1 b	93 ± 7.2 a	4 ± 0.5 a	6.4 ± 0.07 a
Radish							
272	3.7 ± 1.0 a	4.0 ± 1.1 a	1.0 ± 0.2 a	0.7 ± 0.1 a	125 ± 8.1 a	2 ± 0.4 a	6.4 ± 0.09 a
340	4.1 ± 1.0 a	4.4 ± 1.1 a	1.0 ± 0.2 a	0.7 ± 0.1 a	121 ± 9.4 a	2 ± 0.3 a	6.3 ± 0.08 b
408	3.1 ± 0.8 b	3.3 ± 0.9 b	0.9 ± 0.1 b	0.5 ± 0.2 b	144 ± 7.9 a	2 ± 0.3 a	6.5 ± 0.11 a
Green cabbage							
126	2.9 ± 0.2 c	4.9 ± 0.7 c	1.4 ± 0.2 a	0.9 ± 0.2 a	87 ± 9.9 a	3 ± 0.1 b	6.4 ± 0.05 a
180	3.7 ± 0.2 b	3.6 ± 0.6 b	1.1 ± 0.1 b	0.6 ± 0.1 b	87 ± 5.7 a	3 ± 0.1 b	6.4 ± 0.04 a
235	4.4 ± 0.2 a	3.0 ± 0.4 a	1.1 ± 0.1 b	0.5 ± 0.1 b	93 ± 6.9 a	4 ± 0.1 a	6.4 ± 0.03 a
Red cabbage							
126	3.6 ± 0.8 a	4.0 ± 0.9 a	2.9 ± 0.4 c	0.7 ± 0.2 a	94 ± 16.1 b	2 ± 0.3 b	6.1 ± 0.05 b
180	2.9 ± 1.1 a	3.4 ± 1.0 a	3.8 ± 0.6 b	0.6 ± 0.2 a	96 ± 20.0 b	2 ± 0.4 b	6.2 ± 0.09 a
235	3.5 ± 1.2 a	3.9 ± 1.4 a	4.7 ± 0.8 a	0.7 ± 0.3 a	113 ± 25.8 a	3 ± 0.3 a	6.3 ± 0.12 a

[1] Means in the same column that are followed by the same letter do not differ from one another according to Tukey's test ($p > 0.05$).

SS content increased by approximately 1 °Brix at the highest SD for all species except radish. TA was affected only by SD in red cabbage, increasing by 20% at the highest seeding density.

4. Discussion

4.1. Yield and Biometric Characteristics

In the present study, no interaction was observed between PPFD and SD in any of the characteristics analyzed, even on mineral content and phytochemical quality, as further elaborated in the following sections. Given the minimal effect PPFD had, by itself, on those

characteristics, our results indicate that the light intensity range tested was not enough to induce substantial changes in the metabolism and, therefore, was not capable of interacting with SD, which directly affected production across all species investigated.

A positive effect of increasing PPFD in FW and DW has been observed only in the green cabbage experiment. Previous studies have reported a wide range of optimal PPFD levels for microgreen production, highlighting the need to identify the level that balances crop performance with the economic viability of the production system [6,14,19,24,25].

PPFD values between 330 and 440 μmol m^{-2} s^{-1} have been considered optimal for Brassicaceae microgreens, whereas 545 μmol m^{-2} s^{-1} may induce photooxidative stress [14]. However, conflicting results have also been reported, suggesting that lower PPFD levels may be optimal for microgreen cultivation. For instance, 150 μmol m^{-2} s^{-1} was identified as an optimal PPFD for microgreen production [26], while radish microgreens were successfully cultivated under an even lower PPFD of 42 μmol m^{-2} s^{-1} [10], a value close to the lowest treatment evaluated in the present study (50 μmol m^{-2} s^{-1}). This treatment produced the best overall results for sunflower, radish, and red cabbage, as it did not differ from the higher PPFD treatments in terms of yield (FW) while requiring substantially less energy. Similarly, increasing PPFD from 105 to 315 μmol m^{-2} s^{-1} increased mizuna fresh mass by only 15% [24]. Thus, a threefold increase in PPFD resulted in only a modest increase in productivity, indicating lower photon-use efficiency at the higher PPFD.

A similar response was observed in our green cabbage microgreens experiment. Increasing PPFD from 50 to 150 μmol m^{-2} s^{-1} increased fresh weight by only about 10% (approximately 0.4 kg m^{-2}) (Figure 1A). Although substantially more photons were supplied at 150 μmol m^{-2} s^{-1}, the increase in productivity was relatively small. Moreover, the 200 μmol m^{-2} s^{-1} treatment resulted in reduced FW, indicating that this PPFD is not advantageous when maximizing yield is the primary objective.

Since microgreens are marketed as fresh products, their commercial value is directly associated with fresh biomass production, which is therefore the primary parameter used to quantify productivity and determine market value [8]. In contrast, dry mass reflects crop quality by representing the accumulation of carbon and mineral nutrients, which may take the form of various bioactive molecules. Since a PPFD within the range of 50 to 200 μmol m^{-2} s^{-1} had no significant effect on FW or DW for most species evaluated, the use of 50 μmol m^{-2} s^{-1} under a 16 h photoperiod is an effective strategy for microgreen production.

Among the crops commercially produced in indoor farming systems, microgreens require the lowest PPFD because they are harvested at the seedling stage, during which growth is strongly supported by seed reserves. These reserves nourish the embryo until the photosynthetic apparatus becomes fully functional, which occurs only after the onset of the light-dependent phase of development. In the present study, this transition occurred after approximately 40% of the crop cycle had elapsed.

Increasing SD elevated FW in radish and green cabbage up to the highest SDs evaluated. In sunflower and red cabbage microgreens, however, the intermediate SDs produced the highest FW. In sunflower, the highest SD (2125 g m^{-2}) produced an average FW of 7.4 kg m^{-2}, which was 6% lower than the 7.9 kg m^{-2} obtained at 1700 g m^{-2}. This yield exceeded the value reported in previous research, in which 3.7 kg m^{-2} was obtained using an SD of 1696 g m^{-2} [27]. Likewise, a yield of 2.5 kg m^{-2} was reported using an SD of 368 g m^{-2} [28], whereas approximately 0.1 kg m^{-2} was obtained under an SD of 136 g m^{-2} [29].

For red cabbage, no significant difference in FW was observed between 180 and 235 g m^{-2}, indicating that the higher SD was unnecessary. This finding is particularly relevant given the relatively high cost of red cabbage seed on the Brazilian market. In

contrast, radish reached its highest FW at 408 g m^{-2}, closely matching the 4.9 kg m^{-2} reported using an SD of 306 g m^{-2} [28]. Green cabbage also responded positively to the highest SD. These results indicate that excessive increases in SD may either reduce productivity, as observed in sunflower, or result in a productivity plateau, as observed in red cabbage, in which case further increases in SD are not justified. Similarly, a 120% increase in arugula microgreen yield, from 1.2 to 2.6 kg m^{-2}, was reported as SD increased from 14 to 55 g m^{-2} [8].

Overall, increasing SD resulted in FW increases of 20, 47, 51, and 32% for sunflower, radish, green cabbage, and red cabbage microgreens, respectively, corresponding to yield gains of 1.4, 1.7, 1.5, and 0.8 kg m^{-2}. These findings highlight the strong influence of seeding density in microgreen productivity and reinforce the importance of selecting an appropriate SD to achieve high yields while maintaining the economic viability of microgreen production systems.

In the sunflower experiment, hypocotyl length (HL) decreased at the highest seeding density (SD), whereas no such effect was observed in the other species. Cotyledon dimensions, however, decreased as SD increased in all species investigated. HL is an important trait determining harvestability and the commercial quality of microgreens [26,30]. Taller seedlings facilitate both manual and mechanical harvesting, and a hypocotyl length greater than 5 cm is generally desirable [25]. In the present study, all species exceeded this threshold regardless of PPFD. When combined with cotyledon length, total seedling height ranged from approximately 7 to 9 cm. Most species cultivated as microgreens have a total seedling length between 5 and 10 cm, measured from the cut point at the base of the hypocotyl to the apex of the cotyledons [5].

Smaller seeds, such as those of green cabbage and red cabbage, which possess a thin seed coat, exhibited higher yield-to-seed (Y/S) conversion ratios than larger seeds with thick coats, such as sunflower. This is because the seed coat does not contribute to seedling nutrition or fresh weight, although it is included in the total seed weight. Despite the reduction in conversion efficiency as SD increased, productivity increased in all experiments up to a certain seeding density. These results indicate that microgreen productivity is influenced more by the collective contribution of the plant population than by the performance of individual seedlings. Consequently, each species has an optimal seeding density at which yield is maximized.

A low range of light intensity had no effect on the biomass accumulation of sunflower, radish, and green cabbage microgreens, while seed density, an important component of microgreen production, played a major role in determining yield in all species tested. In green cabbage, while a PPFD of 150 µmol m^{-2} s^{-1} increased fresh weight in relation to 50 µmol m^{-2} s^{-1}, the gains were about 10% under a 200% light intensity increase. For comparison, the highest SD resulted in gains of about 52% under an 87% SD increase.

Seeding density outweighed light intensity on the yield of all species investigated, even in the only instance where PPFD had a distinguishable effect. Reducing the energy demands of LED lighting in vertical farming systems is essential for economic feasibility, which can be even further improved by achieving an optimal trade-off between seeding density, seed costs, and fresh yield in microgreens production.

4.2. Mineral Content

Varying light intensity from 50 to 200 µmol m^{-2} s^{-1} influenced none of the sunflower macronutrient content, and few biometric characteristics (Figure 2, Table 2). The lack of PPFD effect in sunflower may be due to its seed morphology, as bigger seeds and consequently cotyledons imply a larger pool of nutrient reserves for germination and initial growth when compared to other species with smaller seeds [31], therefore not reinforcing nutrient demands

at this early stage. Since Brassicaceae seeds are substantially smaller than sunflower seeds, the light stimulus likely had a stronger influence on root-level absorption. Differences in microgreen nutrient content under increased irradiance have also been observed in arugula, a Brassicaceae, though at a larger PPFD range than the one we tested [32].

Generally, a higher nutrient content can enable plants to increase their metabolite pools, which include bioactive compounds like anthocyanins and carotenoids [33]. However, the wide array of signaling pathways in the secondary metabolism of plants results in complex interactions, which are often species-dependent. In our experiments, radish microgreens had, for the most part, the same levels of nutrient content, even though the highest SD resulted in an overall lower phytochemical quality (Table 6). This observation implies that, under a higher number of plants, individual secondary biosynthesis was hindered before primary growth in the form of fresh and dry matter production, which were, in fact, enhanced (Table 1).

Red and green cabbage microgreens were the most sensitive in relation to macronutrient content under different SD, especially K, Ca, and Mg, where their content lowered as SD increased. That stronger response may be related to their nutritional profile and species-specific patterns of mineral accumulation. K is generally the most abundant macronutrient in Brassicaceae microgreens, followed by P, Ca, and Mg, although substantial variation exists among species and cultivars [34]. More recent comparisons among microgreen species have likewise demonstrated considerable genotypic variation in K, Ca, and Mg concentrations, where mineral accumulation is strongly dependent on species [35]. Therefore, the reduction in K, Ca, and Mg concentrations observed with increasing SD should not necessarily be interpreted solely as a reduction in nutrient uptake. Rather, it may reflect changes in the balance between nutrient acquisition and biomass accumulation, particularly as competition increases with plant population density. SD thus represented not only a determinant of microgreen yield but also a factor capable of modifying their nutritional composition, with the magnitude and direction of these responses depending on species and possibly seed reserves.

4.3. Phytochemical Traits

Our findings indicate that, within the range of PPFD levels and species investigated, reducing PPFD does not compromise microgreen phytochemical quality, indicating that 50 μmol m^{-2} s^{-1} was the most efficient PPFD in this regard. Similar findings have been reported for microgreens grown under relatively low PPFD (105 μmol m^{-2} s^{-1}), which exhibited enhanced nutritional quality [24]. This response, however, appears to be species-dependent, as higher irradiance may promote anthocyanin accumulation in red amaranth and carotenoid accumulation in arugula microgreens, though at a wider PPFD range of 280 and 600 μmol m^{-2} $s^{-1,}$ respectively [32,36]. Although increased phytochemical content may be highly desirable from a nutritional and food-quality perspective, the energy input required to achieve such responses should also be considered when designing microgreen cultivation strategies.

As bioactive compounds with antioxidant capacities, molecules such as anthocyanins and carotenoids are associated with benefits to human health, and microgreens are often reported as promising products due to their inherently high quantities of said compounds [23,37]. Even though fresh weight yield is the governing factor in essentially all commercial systems, phytochemical quality may also be considered when defining cultivation parameters such as seed density.

The yield of sunflower microgreens was highest at 1700 g m^{-2} SD (Table 1), which corresponded, for the most part, with a higher phytochemical quality when compared to an SD of 2125 g m^{-2}, except for SS. On the other hand, AN and CA content in green

cabbage microgreens decreased with higher SD, despite the inverse effect SD had on fresh and dry biomass production. This may be attributed to a dilution effect, where constant quantities of a given molecule, combined with an increase in biomass accumulation, lead to lower values per unit of mass [38]. Interestingly, red cabbage microgreens had increased AN values, despite the same fresh yield under SDs of 180 and 235 g m^{-2}, meaning that anthocyanin content differs on a genotype basis in relation to plant density. Red cabbage, as red leafy vegetables in general, tends to have a naturally high AN content [39], evidenced by the fact that the highest AN for green cabbage (1.4 mg 100 g^{-1}) was about three times lower than that of its red counterpart (4.7 mg 100 g^{-1}). In this case, individual plant AN synthesis likely was not strongly affected by a denser environment, even though total fresh weight stayed the same.

5. Conclusions

No interaction was observed between light intensity (PPFD) and seeding density (SD), and in terms of yield, only green cabbage microgreens were responsive to PPFD, though at a level that does not justify a higher irradiance (10% gains in fresh weight yield on a threefold irradiance increase). This means that the lowest PPFD (50 µmol m^{-2} s^{-1}) satisfied the seedling demands and represents the most efficient light intensity management under our experimental conditions.

SD had a pronounced and varied effect on phytochemical quality and nutrient content. Importantly, this effect did not follow the same pattern on yield, which is still the governing factor in microgreens production. Regardless, microgreens can be considered fresh foods with great potential benefits to human health, and our study supports this aspect with the added nuance of SD impacts on their quality.

Overall, SD outweighed PPFD in microgreens production within the tested ranges, where yield improved up to the highest SD in radish and green cabbage (408 and 235 g m^{-2}, respectively), and the second-highest in sunflower and red cabbage (1700 and 180 g m^{-2}). This finding highlights the careful consideration farmers and technicians must give when designing cultivation practices for costly systems like plant factories, as their economic feasibility is still a challenge to healthy food production.

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Abbreviations

The following abbreviations are used in this manuscript:

PPFD	Photosynthetic photon flux density
SD	Seeding density
LED	Light-emitting diode
IAC	Agronomic Institute
NPK	Nitrogen, phosphorus and potassium
EC	Electrical conductivity
FW	Fresh weight
DW	Dry weight
Y/S	Fresh yield-to-seed ratio
CLL	Cotyledon longitudinal length
CTL	Cotyledon transversal length
HL	Hypocotyl length
Chl a	Chlorophyll a
Chl b	Chlorophyll b
AN	Anthocyanins
CA	Carotenoids
TA	Titratable acidity
SS	Soluble solids

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